

Introduction to Digital Signal Processing

HW4: FIR filter design via the window method

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Part I: Hand-written exercises

Exercise 1 (IIR Filter Properties and Frequency Transformation)

1. $H(z) = \frac{B(z)}{A(z)}$ IIR low-pass filter, causal, stable, $y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$

$$\Rightarrow \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

(a) $H(-z) \Big|_{z=e^{j\omega}} = H(-e^{j\omega}) = H(e^{j(\omega-\pi)}) \Rightarrow$ LPF shift π on ω axis, i.e. $H(-z)$ has a high-pass magnitude response

(b) $H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \xrightarrow{(-1)^k = (-1)^k} H(-z) = \frac{\sum_{k=0}^M b_k (-1)^k z^{-k}}{1 + \sum_{k=1}^N a_k (-1)^k z^{-k}}$, i.e. the difference equations of high-pass filter

$y[n] = -\sum_{k=1}^N a'_k y[n-k] + \sum_{k=0}^M b'_k x[n-k]$ with

$$\begin{cases} a'_k = (-1)^k a_k & k=1, \dots, N \\ b'_k = (-1)^k b_k & k=0, \dots, M \end{cases}$$

(c)

cut off frequency: $\frac{1}{5}\pi$ (LPF) $\Rightarrow$ cut off frequency: $\frac{4}{5}\pi$ (HPF)

(d) To design a band-pass filter, we can cascade a LPF and a HPF $\xrightarrow{x[n]} \boxed{\text{LPF}_{\omega_{c1}}} \boxed{\text{HPF}_{\omega_{c2}}} \xrightarrow{y[n]}$

HPF with cut off frequency: ω_{c1} and $\omega_{c1} < \omega_{c2} \Rightarrow$ bandwidth of the filter is $[\omega_{c1}, \omega_{c2}]$

LPF with cut off frequency: ω_{c2} $\rightarrow$ discuss in (0, π)

Figure 1: Derivations for IIR low-pass to high-pass transformation and filter cascading.

Exercise 2 (All-pass Filter and Group Delay)

2. all-pass filter $H(z) = \frac{1 - qz^{-1}}{1 - pz^{-1}}$, $p = re^{j\theta}$, $q = \frac{1}{p^*} = \frac{1}{r}e^{j\theta}$, $r < 1$, $H(z)$: causal, stable

(a) $Z(s) = \frac{1+s}{1-s}$, $H_c(s) = H(Z(s)) = \frac{\left(\frac{1+s}{1-s}\right) - q}{\left(\frac{1+s}{1-s}\right) - p} = \frac{\left(\frac{1+q}{1+p}\right) \cdot \frac{s - \frac{q-1}{q+1}}{s - \frac{p-1}{p+1}}}{1}$

$\downarrow$

$S(z) = \frac{z-1}{z+1}$ $\leftarrow \begin{array}{l} \text{zero} \leftrightarrow \text{zero} \\ \text{pole} \leftrightarrow \text{pole} \end{array}$

$S_p = \frac{p-1}{p+1} = \frac{(re^{j\theta} - 1)(re^{-j\theta} + 1)}{(re^{j\theta} + 1)(re^{-j\theta} + 1)} = \frac{(r^2 - 1) + j(2r \sin \theta)}{1 + |p|^2} = u_p + jv_p$

$q = \frac{1}{p^*}$

$S_q = \frac{q-1}{q+1} = \frac{1 - p^*}{p^* + 1} = \frac{(1 - re^{-j\theta})(re^{j\theta} + 1)}{(re^{-j\theta} + 1)(re^{j\theta} + 1)} = \frac{(1 - r^2) + j(2r \sin \theta)}{1 + |p|^2} = u_q + jv_q$

$\begin{cases} u_p = \frac{r^2 - 1}{1 + |p|^2} < 0 \because r < 1 \\ v_p = \frac{2r \sin \theta}{1 + |p|^2} \\ u_q = \frac{1 - r^2}{1 + |p|^2} > 0 \because r < 1 \\ v_q = \frac{2r \sin \theta}{1 + |p|^2} \end{cases}$

$\therefore u_p = -u_q < 0$, and $v_p = v_q$

(b) From above, we have $\angle H_c(j\Omega) = \angle \left(\frac{1+q}{1+p} \right) + \underbrace{\angle(j\Omega - S_q)}_{\text{constant}} - \angle(j\Omega - S_p)$ W.L.O.G. $\sin \theta > 0$

S -plane

$\Omega \uparrow \Rightarrow \begin{cases} \theta_p \uparrow \rightarrow \frac{\partial \theta_p}{\partial \Omega} > 0 \\ \theta_q \downarrow \rightarrow \frac{\partial \theta_q}{\partial \Omega} < 0 \end{cases}$

$\frac{\partial \angle H_c(j\Omega)}{\partial \Omega} = \frac{\partial \theta_q}{\partial \Omega} - \frac{\partial \theta_p}{\partial \Omega} < 0, \forall \Omega$

$\frac{\partial \theta_q}{\partial \Omega} < 0, \frac{\partial \theta_p}{\partial \Omega} > 0$

(c) $\Omega = \tan\left(\frac{\omega}{2}\right)$, $\frac{\partial \angle H(e^{j\omega})}{\partial \omega} = \frac{\partial \angle H_c(j\Omega)}{\partial \Omega} \cdot \left(\frac{d\Omega}{d\omega}\right) < 0, \forall \omega \Rightarrow$ the filter has a positive group delay

$< 0 \text{ (b)} > 0$

Figure 2: Analysis of All-pass filter stability and phase response properties.

Part II: Computer-based exercises

Exercise 3 (*FIR Filter Design via Window Method*)

Design Standards:

The target FIR filter must satisfy: $\omega_p = \pi/3$, $\omega_s = 0.4\pi$, stopband gain $|H(e^{j\omega_s})| \leq 0.01$ (-40 dB), and passband ripple ≤ 0.01 .

(a) Minimum Filter Length via Hann Window

We selected the Hann window because its peak side-lobe is -44 dB, meeting our -40 dB requirement. The design was implemented as follows:

```
win = hann(order + 1); % Stores Hann window weights (0 to 1)
b = fir1(order, omega_cut/pi, 'low', win); % Stores impulse
    response coefficients
```

The frequency response was verified using the DTFT definition: $H(e^{j\omega}) = \sum_{n=0}^M b_n e^{-j\omega n}$. By iterative search, we found the minimum order $M = 92$ (shown as $N = 46$ in the plot) to meet the specs.

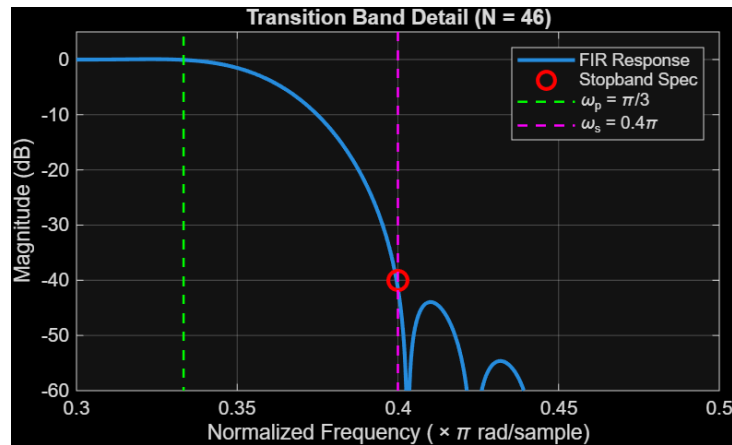


Figure 3: Transition band detail for $N = 46$, satisfying the -40 dB attenuation at 0.4π .

(b) Impact of ω_s on Filter Order N

We fixed $\omega_p = \pi/3$ and varied $\omega_s \in [0.4\pi, 0.5\pi]$ to observe the minimum required order N .

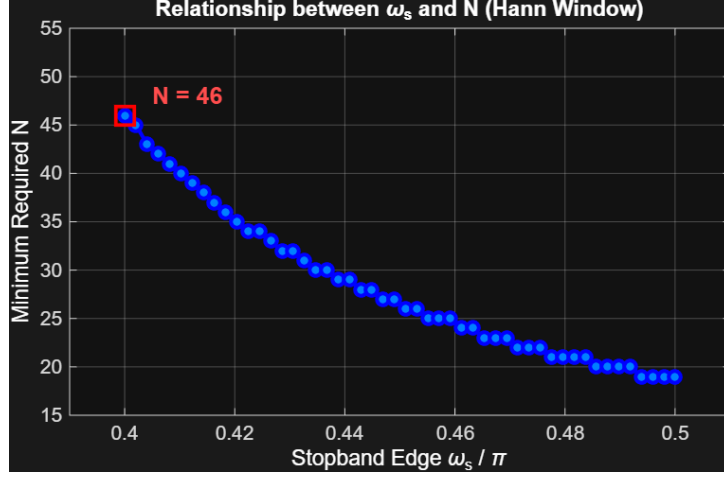


Figure 4: Minimum required order N versus stopband edge ω_s .

- **Inverse Relationship:** N decreases non-linearly as ω_s increases, confirming that transition bandwidth $\Delta\omega$ is inversely proportional to filter length M .
- **Staircase Effect:** The discrete integer nature of N results in the observed staircase plot.
- **Result:** At $\omega_s = 0.4\pi$, $N = 46$ is required, but this requirement relaxes as ω_s moves further from ω_p .

(c) Performance Comparison: Butterworth (IIR) vs. Hann (FIR)

We compared the 21st-order Butterworth filter with the 92th-order Hann-window FIR filter, both designed for $\omega_s = 0.4\pi$.

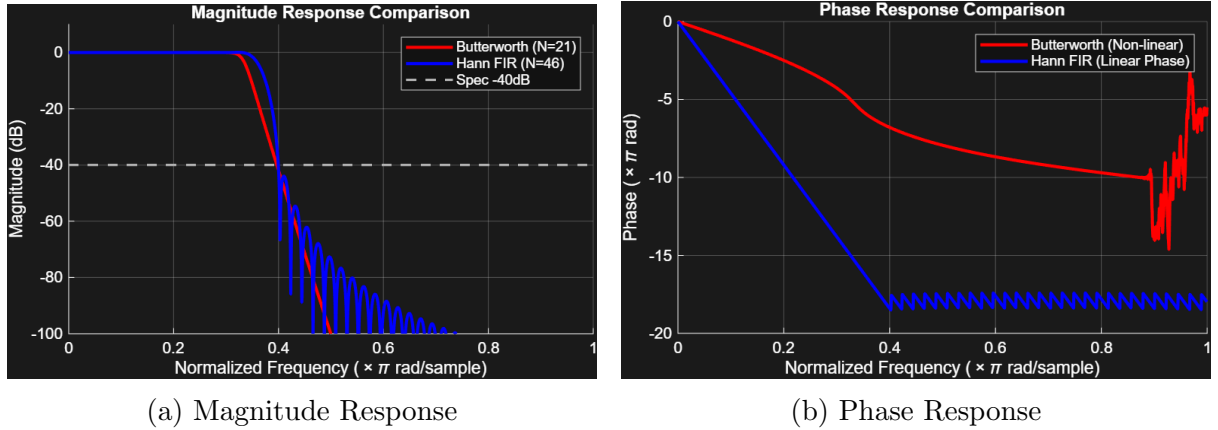


Figure 5: Frequency response comparison between IIR and FIR designs at $\omega_s = 0.4\pi$.

- **Efficiency:** Butterworth is more efficient, using fewer coefficients ($N = 21$) than FIR ($N = 46$, $M = 92$) for the same -40 dB spec.
- **Phase Linearity:** FIR provides a perfectly linear phase (straight line), avoiding the phase distortion seen in the non-linear Butterworth filter.
- **Roll-off:** Butterworth achieves a sharper transition with lower complexity, while FIR offers stability and phase consistency at the cost of higher computation.